

Lab 3: Hello, CUDA!

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Each CPU iteration and each GPU thread computes $Y = 0.299R + 0.587G + 0.114B$. The RGB image is flattened to $(HW, 3)$ for the 1D kernel. Blocks of 32–512 threads are compared. The notebook shows both images, checks the maximum pixel error, and plots block size versus GPU time. Select the fastest measured block; larger blocks do not necessarily run faster.

Kernel timing uses `time.time()` after JIT warm-up and CUDA synchronization, averaged over three GPU runs. Device allocation and host/device copies are excluded. CPU measurements are single runs. Speedup is relative to the stated CPU implementation. Actual measurements appear after running the notebook on Colab; no timings are assumed here.